

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1153

Southampton Solar Llc 100 kV
June 28, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1153. The Interconnection Customer proposes a 151.0 MW wind facility interconnecting at the Southampton Solar Llc 100 kV substation in the Ohio area. The request was received on February 18, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 394 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,177,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Nordonia / Hills 100 kV L is overloaded to 197.5% of its 157 MVA emergency rating for the loss of the Imperial / Chamberlin 230 kV L. The project contributes 8.9% DFAX (13.4 MW) to this overload.

For the loss of Chamberlin / Imperial 230 kV, the Lincoln Beach / Hills 100 kV L loads to 157.7% of its 127 MVA rating. Project DFAX: 10.2%.

For the loss of Sranton / Marcos 100 kV L, the Cleveland Thermal / Marcos 100 kV loads to 148.9% of its 130 MVA rating. Project DFAX: 6.4%.

For the loss of Imperial / Chamberlin 230 kV line, the Nordonia / Chamberlin 100 kV loads to 133.3% of its 163 MVA rating. Project DFAX: 7.1%.

The Scranton / Marcos 100 kV is overloaded to 128.9% of its 130 MVA emergency rating for the loss of the Cleveland Thermal / Marcos 100 kV. The project contributes 5.4% DFAX (8.1 MW) to this overload.

For the loss of Imperial / Lincoln Beach 100 kV, the Imperial / Lincoln Beach 100 kV Ckt 1 loads to 123.6% of its 152 MVA rating. Project DFAX: 8.0%.

The Imperial / Lincoln Beach 100 kV #2 is overloaded to 121.8% of its 154 MVA emergency rating for the loss of the Imperial / Lincoln Beach 100 kV Line 1. The project contributes 8.0% DFAX (12.1 MW) to this overload.

For the loss of Imperial / Eufaula 500 kV, the W 41st Street / Loring Power Plant 100 kV loads to 114.2% of its 112 MVA rating. Project DFAX: 12.3%.

The Bath / Tyco Electronics 100 kV is overloaded to 106.6% of its 122 MVA emergency rating for the loss of the Nordonia / Hills 100 kV L. The project contributes 5.0% DFAX (7.6 MW) to this overload.

The Bthel / Okeelanta Cogeneration 100 kV L is overloaded to 104.4% of its 133 MVA emergency rating for the loss of the Hylebos / Ravenna 500 kV. The project contributes 5.5% DFAX (8.3 MW) to this overload.

The Dow St Charles Operations / Bath 100 kV Line is overloaded to 103.8% of its 113 MVA emergency rating for the loss of the Hills / Nordonia 100 kV. The project contributes 5.0% DFAX (7.6 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hills / Nordonia 100 kV	Rebuild line with higher-capacity conductor	\$3,425,000	14% (\$493,000)
Hills / Lincoln Beach 100 kV	Rebuild line with higher-capacity conductor	\$2,985,000	12% (\$367,000)
Cleveland Thermal / Marcos 100 kV	Rebuild line with higher-capacity conductor	\$1,127,000	15% (\$174,000)
Chamberlin Substation / Nordonia 100 kV line	Rebuild line with higher-capacity conductor	\$2,432,000	15% (\$368,000)
Marcos / Scranton 100 kV	Rebuild line with higher-capacity conductor	\$1,962,000	17% (\$329,000)
Imperial / Lincoln Beach 100 kV Ckt 1	Rebuild line with higher-capacity conductor	\$1,219,000	17% (\$210,000)
Lincoln Beach / Imperial 100 kV Circuit 2	Rebuild line with higher-capacity conductor	\$1,421,000	17% (\$245,000)
West 41St Street / Loring Power Plant 100 kV	Rebuild line with higher-capacity conductor	\$5,357,000	61% (\$3,245,000)
Bath Substation / Tyco Electronics 100 kV	Reconductor line	\$802,000	19% (\$152,000)
Bthel / Okeelanta Cogeneration 100 kV L	Reconductor line	\$418,000	100% (\$418,000)
Do St Charles Operations / Bath Substation 100 kV	Reconductor line	\$929,000	19% (\$176,000)
	Total allocated to project		\$6,177,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.